

B.Tech III Year II Semester (R20) Supplementary Examinations November/December 2024

OPTICAL COMMUNICATIONS

(Electronics & Communication Engineering)

Time: 3 hours

Max. Marks: 70

PART – A

(Compulsory Question)

1 Answer the following: (10 X 02 = 20 Marks)

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| (a) Write the advantages of optical fiber communications. | 2M |
| (b) What is total internal reflection? | 2M |
| (c) Define the return loss with respect to an optical connector. | 2M |
| (d) What is microscopic optical bending? | 2M |
| (e) Define internal quantum efficiency of LED. | 2M |
| (f) What is fiber splicing? | 2M |
| (g) What is the error sources associated with fiber optic receiver section? | 2M |
| (h) Define the Probability of error. | 2M |
| (i) List out the various design considerations of an optical system. | 2M |
| (j) Define Rise time. | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

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| 2 (a) Define Numerical aperture and derive the expression for it. | 5M |
| (b) Briefly explain about Graded Index fibrers. | 5M |

OR

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| 3 (a) Explain briefly the difference between single-mode and multimode fiber. Which one gives better performance and why. | 5M |
| (b) An optical fiber cable has values of refractive index of 1.46 for the fiber core. If the fractional change in the index of refraction of the optical fiber is specified as 0.01, then find its numerical aperture. | 5M |

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| 4 (a) Differentiate between Intra modal and intermodal types of dispersion. What are the components of Intra modal dispersion in a single-mode optical fiber? | 5M |
| (b) Describe the various types of fiber optic connectors. | 5M |

OR

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| 5 (a) What are different types of transmission losses that an optical signal incurs when it propagates through the optical fiber cable? Explain any two types of losses. | 5M |
| (b) What are the various types of dispersion in optical fibers? Explain the intermodal dispersion in a step-index fiber. | 5M |

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| 6 (a) Draw a cross-section of an LED structure, and describe its operation. | 5M |
| (b) Explain the v-groove fiber splicing technique of optical fibers with a neat diagram. | 5M |

OR**Contd. in Page 2**

- 7 (a) Explain the elastic-tube fiber splicing technique of optical fibers with a neat diagram. 5M
(b) Construct the Fabry-Perot resonator cavity laser diode with necessary diagram also derive the threshold condition for lasing. 5M
- 8 (a) Discuss the advantages of the structures used for PIN photodiode over that used for p-n photodiode. 5M
(b) Explain the operation of various basic blocks of an optical receiver with a neat diagram. 5M
- OR**
- 9 (a) Explain the operation of PIN photodiode with a neat diagram and describe its energy band structure. 5M
(b) Explain the various error sources in an optical receiver. 5M
- 10 (a) How WDM technique is different from FDM technique? List various advantages of WDM technique. 5M
(b) Explain in detail the various steps in the measurement of dispersion of an optical link. 5M
- OR**
- 11 (a) Explain the principle of WDM with a neat diagram. 5M
(b) Explain in detail the various steps in the measurement of attenuation of an optical link. 5M
